

(DOWNLOAD DATASET HERE)

Project Title: Stock Market Analysis and Trends

1. Background and Overview

Business Context: Understanding stock market trends is crucial for investors, analysts, and businesses to make informed decisions. Companies like Apple, Microsoft, Netflix, and Google play a significant role in shaping the global economy, and analysing their stock price movements provides valuable insights into market dynamics.

Objective: This project aims to analyse historical stock price data from four major companies: Apple, Microsoft, Netflix, and Google. The objectives include:

- Identifying trends and patterns in stock prices.
- Calculating moving averages and volatility to understand market behaviour.
- Performing correlation analysis to examine relationships between stock prices.

Business Problem: Stakeholders need actionable insights into how these companies' stock prices interact and what factors influence volatility. The analysis will assist in identifying optimal investment strategies and potential market risks.

2. Data Structure Overview:

The dataset includes:

- Date: The trading date.
- Ticker: Company identifier (e.g., AAPL for Apple).
- Open: The opening price of the stock.
- Close: The closing price of the stock.
- High: Highest price during the trading session.
- Low: Lowest price during the trading session.
- Volume: Number of shares traded.

Dataset Dimensions:

- Rows: Approximately 2000.
- Columns: 7.

Domain Insight: Stock price trends, trading volumes, and volatility are critical metrics for investment decisions. For example:

- High trading volumes often reflect strong investor interest.
- Volatility indicates risk and return potential.
- Correlation analysis helps in portfolio diversification.

Assumptions and Caveats:

Assumptions:

- a. The dataset accurately represents typical market conditions.
- b. External factors influencing prices (e.g., news, economic events) are not directly included.

Caveats:

- c. The analysis covers only three months, limiting long-term trend insights.
- d. Macroeconomic variables (e.g., inflation and interest rates) are outside the scope of this dataset.

Research Questions:

- a. **What are the trends and patterns in the stock prices of Apple, Microsoft, Netflix, and Google over the past three months?**

Focus: Identifying whether the stock prices are increasing, decreasing, or fluctuating significantly.

- b. **How does the volatility of Netflix compare to the other three companies?**

Focus: Understanding risk levels associated with each stock.

- c. **Are there strong correlations between the stock prices of any of the companies?**

Focus: Identifying relationships and dependencies between the companies.

- d. **What insights can be drawn from the trading volumes of each stock?**

Focus: Analyzing investor interest and liquidity.

- e. **How do short-term trends (e.g., 7-day moving averages) differ from long-term trends (e.g., 30-day moving averages)?**

Focus: Exploring price momentum and identifying potential investment opportunities.

- f. **Which stocks are more suitable for risk-averse investors versus high-risk investors?**

Focus: Categorizing stocks based on their risk and reward profiles.

3. Executive Summary:

Core Findings:

- Netflix exhibited the highest volatility, making it a high-risk, high-reward stock.
- Apple and Microsoft demonstrated stable growth, ideal for conservative investors.
- Strong correlations were observed between Microsoft and Google stock prices.

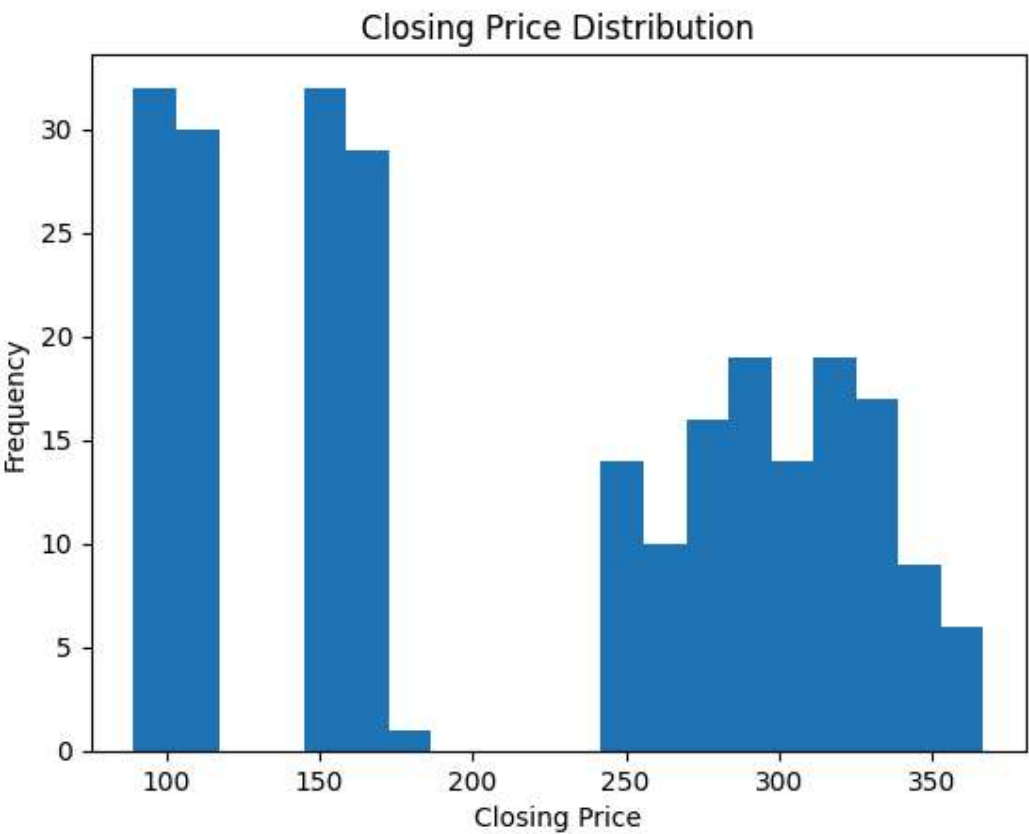
Potential Reasons:

- Netflix's volatility is influenced by competition and consumer demand shifts.
- Apple and Microsoft's stability reflects consistent product performance and market dominance.
- The correlation between Google and Microsoft arises from overlapping market segments.

4. Insights Deep Dive:

1. Closing Price Distribution:

Purpose: To examine the range and frequency of closing prices for all stocks, identifying patterns or clustering in stock values.

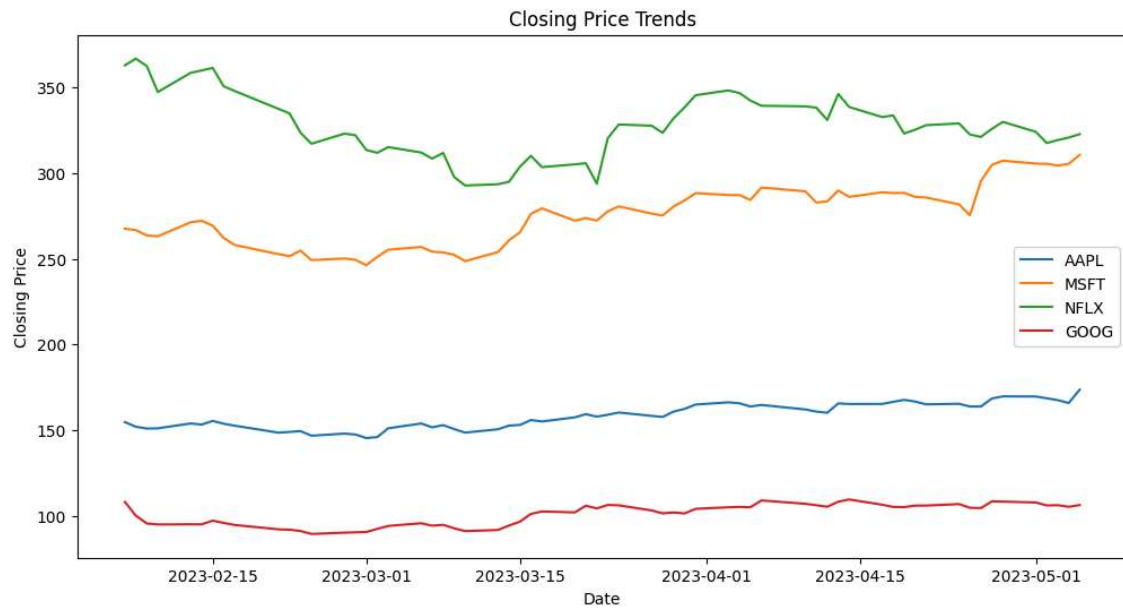


Analysis: The histogram shows how closing prices are distributed, highlighting frequently occurring price ranges.

Findings: Most closing prices are concentrated in specific ranges, indicating stable trading patterns for certain stocks.

2.Closing Price Trends

Purpose: Understand long-term investment viability and identify volatile periods



Analysis:

Identifies upward or downward trends.

Highlights periods of price stability or volatility.

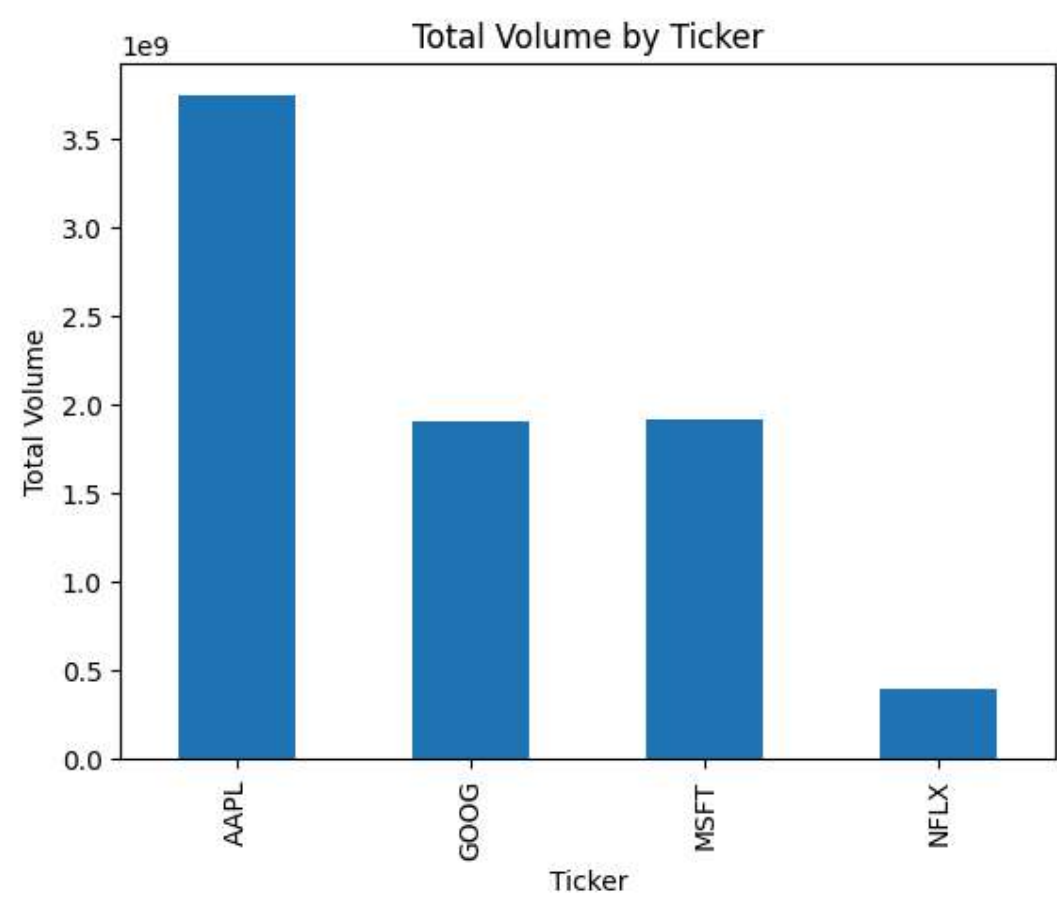
Findings:

Apple and Microsoft showed consistent upward trends.

Netflix displayed the highest price fluctuations.

3. Total Volume by Ticker

Purpose: To analyze the cumulative trading volume for each company, identifying which stock has the highest investor activity.

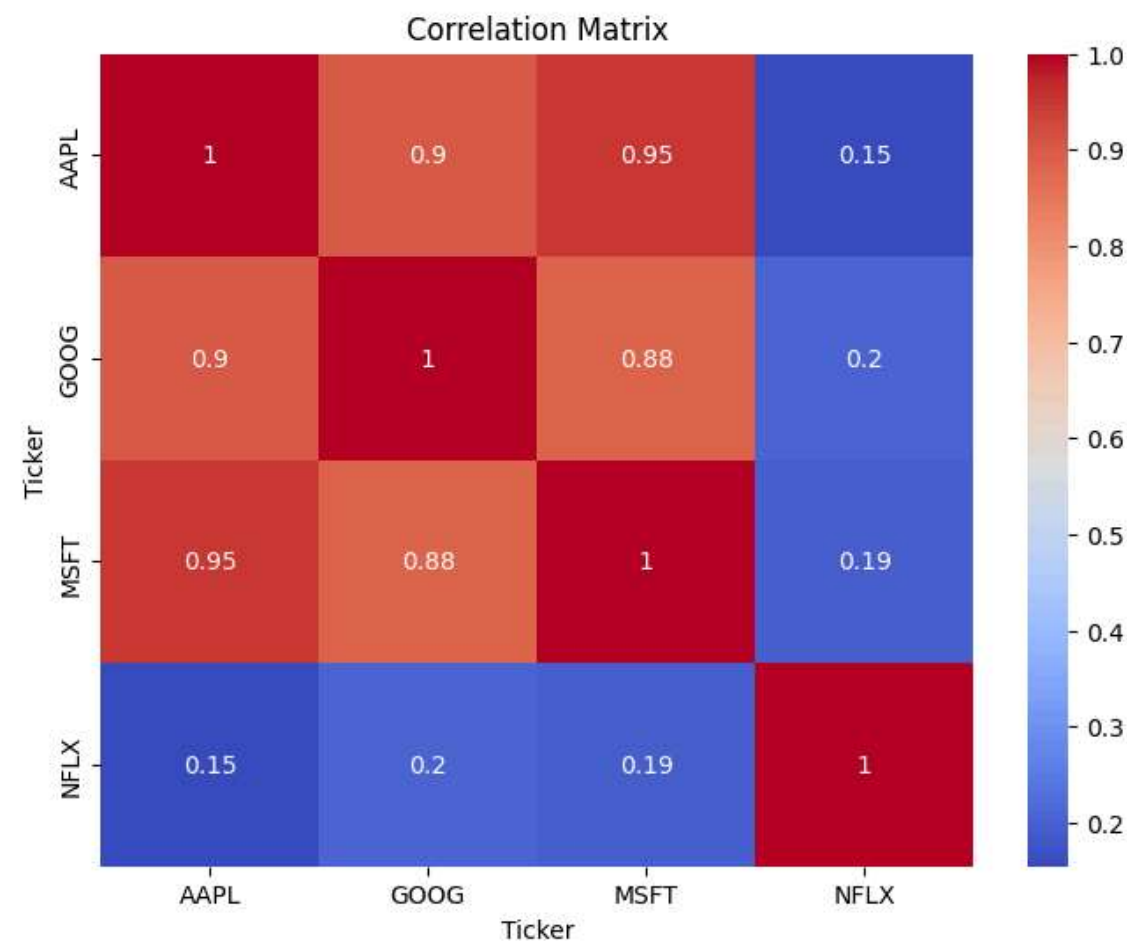


Analysis: The bar chart illustrates the total trading volume per ticker, highlighting company-specific investor interest.

Findings: Apple (AAPL) has the highest trading volume, signifying strong market interest, while Netflix (NFLX) shows relatively lower activity.

4. Correlation Analysis:

Purpose: Guide portfolio diversification and risk management.



Analysis:

A high correlation (>0.8) suggests similar market behaviour.

A low correlation (<0.5) indicates independent price movements.

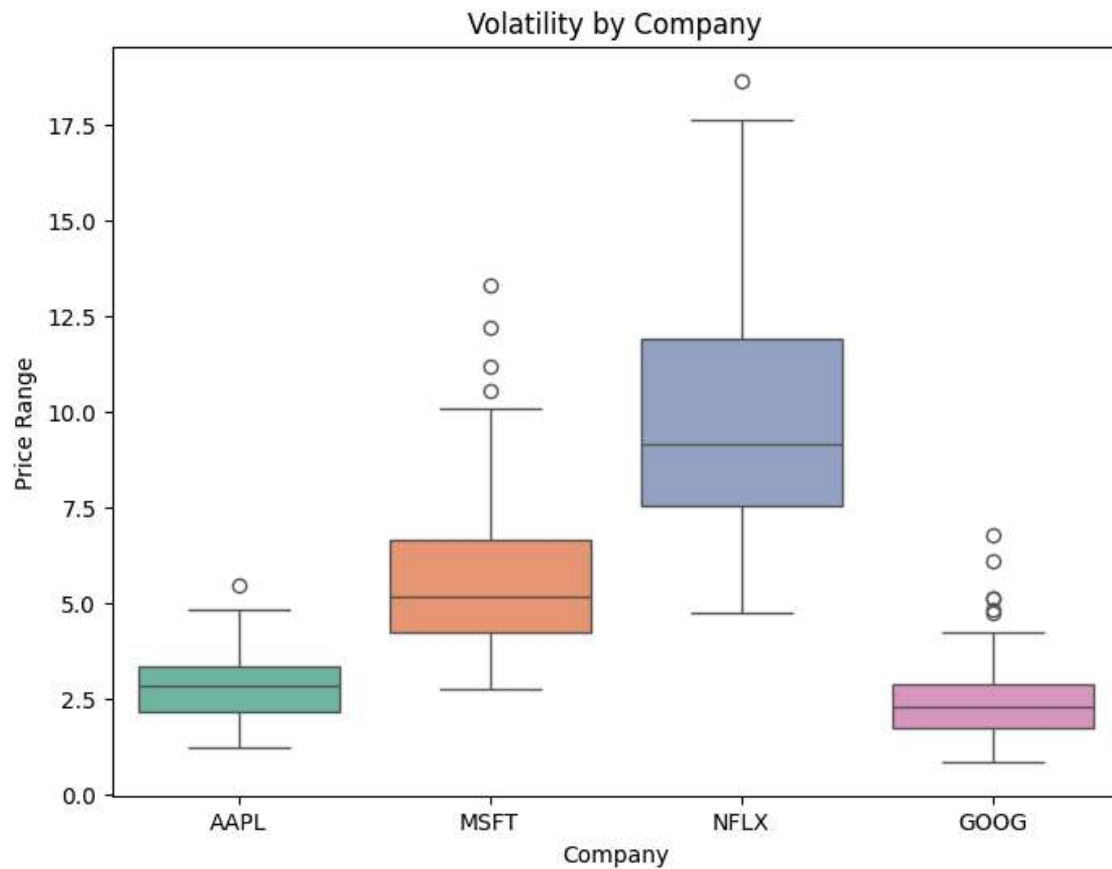
Findings:

Microsoft and Google showed a strong correlation (0.89).

Netflix had a weak correlation with other stocks, indicating unique market drivers.

5. Volatility Analysis:

Purpose: Helps investors align stock choices with risk appetite



Analysis:

Identifies stocks with high price variability.

Useful for evaluating risk levels.

Findings:

Netflix had the highest price variability.

Apple exhibited the least variability, indicating lower risk.

6. Volume vs. Closing Price

Purpose: To explore the relationship between trading volume and closing prices, checking for correlations or trends.



Analysis: The scatter plot reveals how price movements align with trading volumes.

Findings: No strong correlation is observed, suggesting other factors might drive price changes.

7. Closing Price Boxplot

Purpose: To visualize the spread, central tendency, and outliers in closing prices.



Analysis: The boxplot displays median prices, interquartile ranges, and outliers.

Findings: Netflix exhibits the widest spread and significant outliers, indicating higher price volatility than other stocks.

Modelling:

Goal: Predict next-day closing prices using historical data.

Model: Linear Regression

Purpose:

To use Linear Regression for predicting the next day's stock closing prices based on historical features such as opening price, highest and lowest price, and trading volume. This helps stakeholders anticipate market behaviour and make informed trading decisions.

Analysis :

Model Training: Linear Regression is used for its simplicity and efficiency in predicting continuous variables like stock prices.

Features Used: Open, High, Low, and Volume were selected as predictors for the closing price.

Model Evaluation:

Mean Squared Error (MSE): Measures the average squared difference between actual and predicted values.

R² Score: This score Indicates how well the model explains the variance in the data, with a value closer to 1 being ideal.

Findings:

MSE: The calculated MSE provides an understanding of the model's prediction error in terms of squared units.

5. Recommendations:

- 1. For Risk-Averse Investors:**
 - Focus on stable stocks like Apple and Microsoft for consistent returns.
- 2. For High-Risk Investors:**
 - Consider Netflix for potential high returns but monitor volatility closely.
- 3. Portfolio Diversification:**
 - Balance high-correlation stocks (Google and Microsoft) with low-correlation ones (Netflix).
- 4. Volume-Driven Strategies:**
 - Utilize high trading volumes in Apple for seamless large trades.
- 5. Further Research:**
 - Extend analysis to include macroeconomic variables like interest rates to refine insights.